repairs, while simultaneously adjusting blade pitch across the farm to maximize energy capture based on real-time weather patterns.

1.2.1 Internet of things (IoT)

Not too long ago, measuring things such as temperature, vibration, location, or air quality required bulky and expensive equipment. The affordable and compact sensors available today enable users to capture detailed data about virtually any physical environment or system, opening unprecedented possibilities for monitoring and control. Hobbyists can now build sophisticated home automation systems that measure everything from air quality to energy usage, while startups can deploy sensor networks to monitor equipment performance or environmental conditions at a fraction of historical costs.

> ***NOTE***
>
> The DHT22 sensor that I use to measure temperature and humidity in my home costs around $5 today. Three decades ago, comparable digital measurement required industrial sensors and data acquisition systems costing thousands of dollars, accessible only to well-funded organizations.

Large enterprises leverage these same technologies to enable monitoring of manufacturing lines, predictive maintenance of critical equipment, and optimization of resource usage. The sensors already embedded in smartphones demonstrate the power of this technology, with users able to measure motion, orientation, light levels, and even perform 3D spatial mapping, turning everyday devices into powerful data collection tools. Figure 1.1 shows output from an iPhone's three-axis accelerometer while the phone is carried in a pocket. The distinctive pattern in this data allows us to infer the physical state of the system, namely that the person carrying it is walking. This democratization of sensing technology means that anyone can now gather the rich, continuous data streams necessary to build digital twins that were once only available to organizations with massive budgets and specialized expertise.